

Dynamic Quantization Range Control for Analog-in-Memory Neural Networks Acceleration

NATHAN LAUBEUF, [Interuniversity Microelectronics Center](#) (IMEC), Belgium and ESAT Laboratory, [Katholieke Universiteit](#) (K.U.) Leuven, Belgium
JONAS DOEVENSPECK, IOANNIS A. PAPISTAS, MICHELE CASELLI, STEFAN COSEMANS, PETER VRANCX, DEBJYOTI BHATTACHARJEE, ARINDAM MALLIK, PETER DEBACKER, and DIEDERIK VERKEST, [Interuniversity Microelectronics Center](#) (IMEC), Belgium
FRANCKY CATTHOOR and RUDY LAUWEREINS, [Interuniversity Microelectronics Center](#) (IMEC), Belgium and ESAT Laboratory, [Katholieke Universiteit](#) (K.U.) Leuven, Belgium

Analog in Memory Computing (AiMC) based neural network acceleration is a promising solution to increase the energy efficiency of deep neural networks deployment. However, the quantization requirements of these analog systems are not compatible with state-of-the-art neural network quantization techniques. Indeed, while the quantization of the weights and activations is considered by modern deep neural network quantization techniques, AiMC accelerators also impose the quantization of each **Matrix Vector Multiplication (MVM)** result. In most demonstrated AiMC implementations, the quantization range of MVM results is considered a fixed parameter of the accelerator. This work demonstrates that dynamic control over this quantization range is possible but also desirable for analog neural networks acceleration. An AiMC compatible quantization flow coupled with a hardware aware quantization range driving technique is introduced to fully exploit these dynamic ranges. Using CIFAR-10 and ImageNet as benchmarks, the proposed solution results in networks that are both more accurate and more robust to the inherent vulnerability of analog circuits than fixed quantization range based approaches.

CCS Concepts: • **Computing methodologies** → **Neural networks**; • **Hardware** → **Analog and mixed-signal circuits**;

Additional Key Words and Phrases: Neural networks, quantization, in-memory-computing

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Author's current address: J. Doevenspeck, I. A. Papistas, and S. Cosemans, Axelera AI, Sluisstraat 79, Leuven, Belgium, 3000; emails: {jonas.doevenspeck, ioannis.papistas, stefan.cosemans}@axelera.ai; M. Caselli, Department of Engineering and Architecture, University of Parma, Parco Area delle Scienze, 181/A, Parma, Italy, 43124; email: michele.caselli@unipr.it. Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than ACM must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

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